Project Presentation

Kasetsart University

A Comparison of the Efficiency for Predicting House Price in Bangkok Metropolitan

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Project Pre... Members

Group Members

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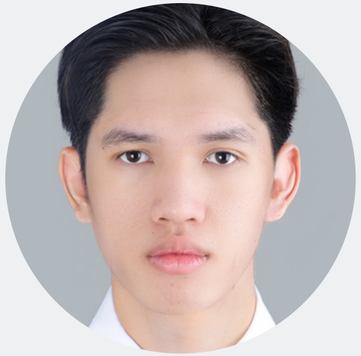

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Introduction

The real estate market in Bangkok and its metropolitan area remains one of the largest and most important sectors contributing to Thailand economy. The demand for residential housing continues to grow, Reflecting changes in consumer behavior influenced by demographic structures, lifestyles, expectations regarding quality of life, value, and environmental safety. However, several economic, social, and environmental factors have increased the complexity of the housing market. These include rising land prices, increasing construction costs, and psychological factors affecting consumer decision-making. Furthermore, recent natural disasters, such as earthquakes, have affected consumer confidence in high-rise residential buildings, leading some buyers to shift their preferences toward low-rise housing options.

Traditionally, house price prediction has relied on classical statistical methods such as linear regression. While these approaches can explain relationships between variables to some extent, they often have limitations when handling nonlinear relationships or complex interactions among multiple variables. With recent technological advancements, machine learning techniques have become increasingly important tools for improving prediction accuracy, as they can effectively learn from large datasets and detect complex patterns. Therefore, this study aims to compare the predictive performance of traditional statistical methods and machine learning techniques to identify the most effective model for house price prediction in the Bangkok metropolitan region.

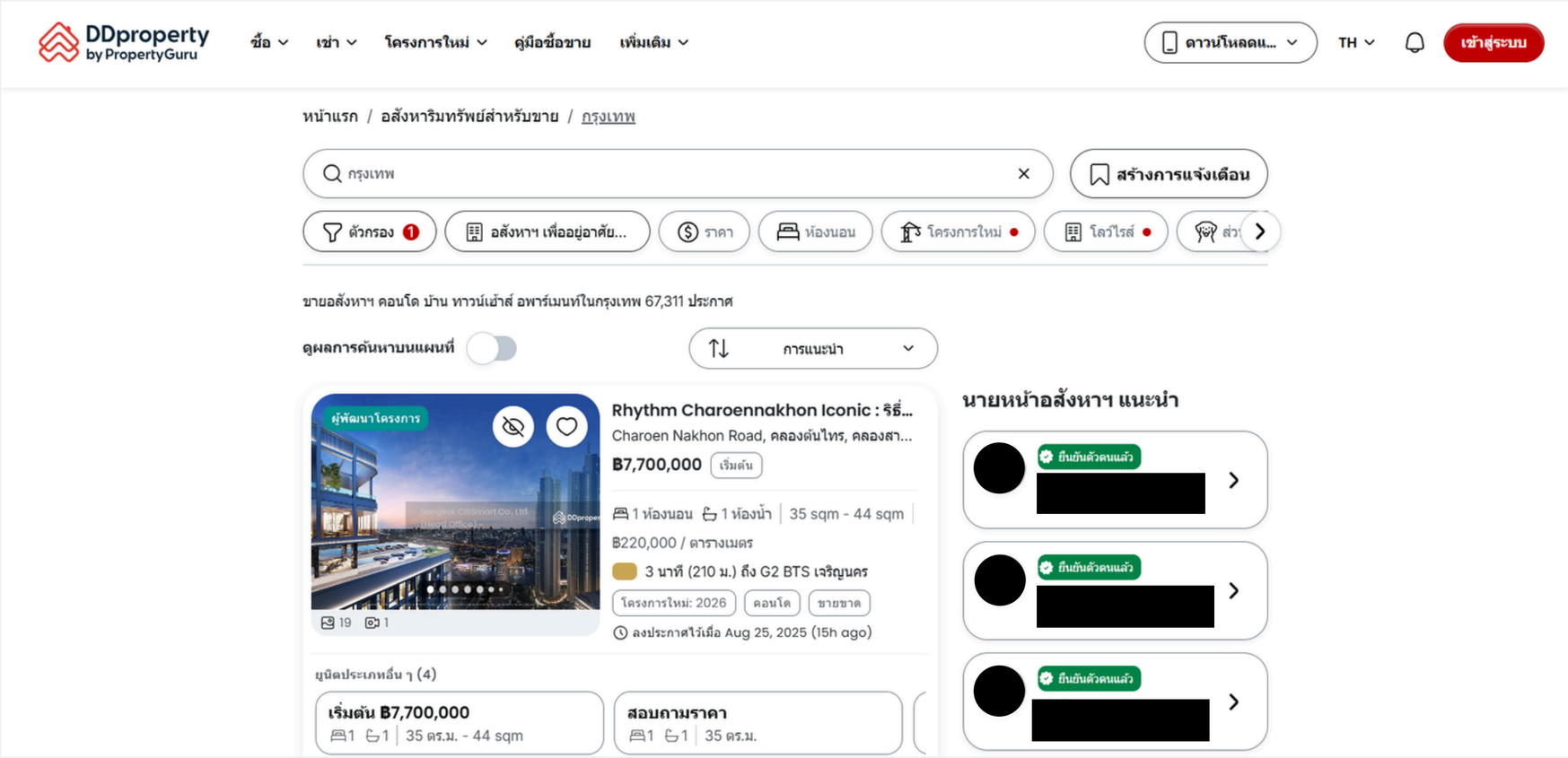
Objectives

- Compare the performance of Linear Regression, Random Forest, ANN, and XGBoost
- Identify key factors affecting housing prices in Bangkok metropolitan area

Hypothesis

- XGBoost achieves the best performance in house price prediction.
- Location is the most influential factor affecting house prices.

Population and Sample



Variables

Variables	Label	Measurement scale	Type / Unit
Y	house price	numerical	million THB
X1	house type	categorical	detached house, townhouse
X2	house district	categorical	inner, middle, outer, metropolitan
X3	house size	numerical	square meters (m2)
X4	number of bedrooms	numerical	rooms
X5	number of bathrooms	numerical	rooms

Variables

Variables	Label	Measurement scale	Type / Unit
X6	distance to the nearest school	numerical	kilometer (km)
X7	distance to the nearest shopping mall	numerical	kilometer (km)
X8	distance to the nearest healthcare	numerical	kilometer (km)
X9	distance to the nearest park	numerical	kilometer (km)
X10	distance to the nearest train station	numerical	kilometer (km)

Procedure

DATA COLLECTION

Data were collected from the DDproperty website using a web scraping technique through the Web Scraper tool. A total of 1,000 residential property records were obtained. The dataset consists of 11 variables, including one dependent variable (house price) and ten independent variables related to housing characteristics such as house type, house size, and location factors.

DATA MANAGEMENT

Data preprocessing was performed to improve data quality, including data cleaning, handling missing values, and transforming categorical variables into suitable formats. After preprocessing, variable selection was conducted using the Stepwise method in SPSS to identify the most significant predictors.

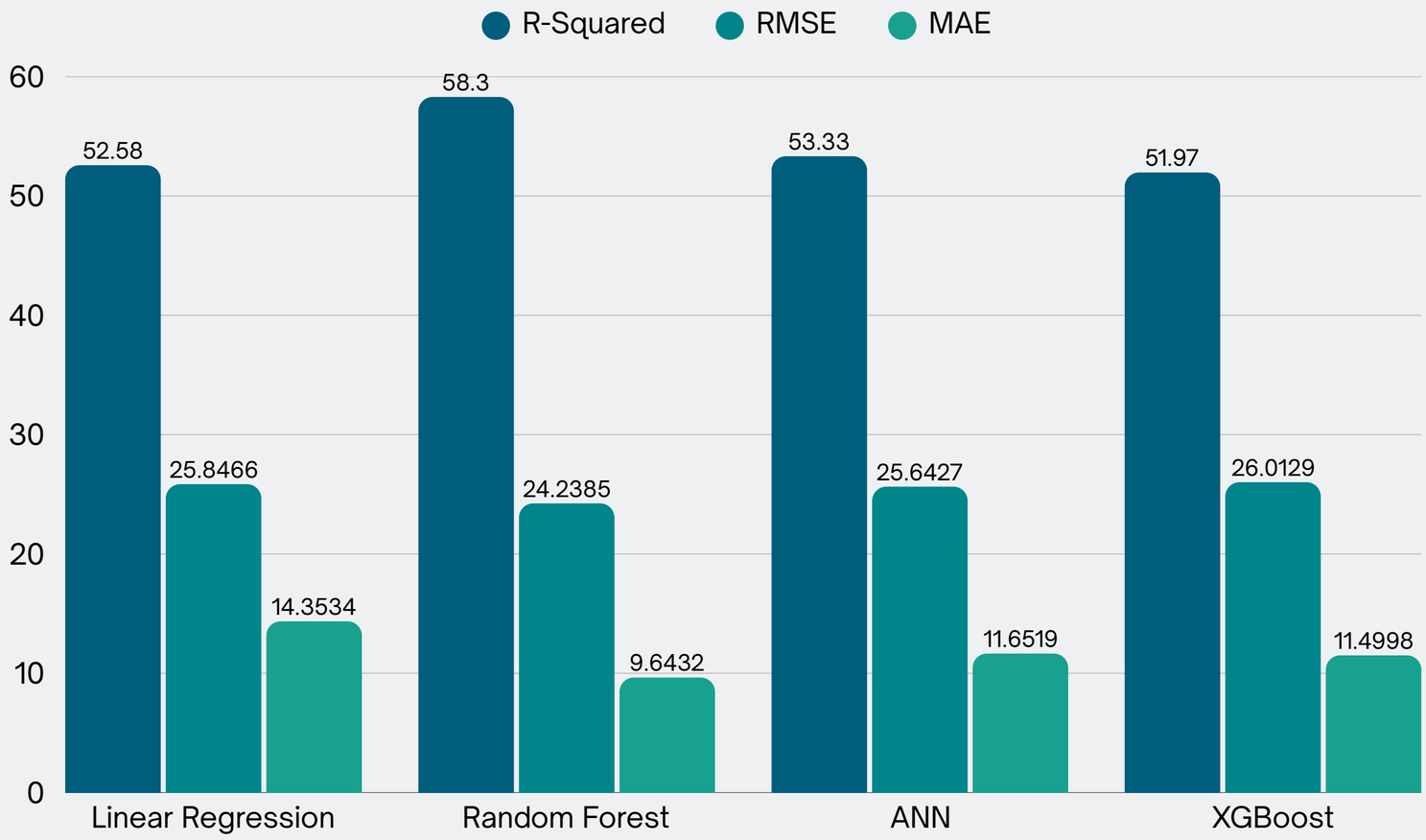
MODEL DEVELOPMENT&EVALUATION

The dataset was divided into training (70%) and testing (30%) sets. Four models (Linear Regression, Random Forest, ANN, and XGBoost) were developed using the same dataset to ensure a fair comparison. Model performance was evaluated using R^2, RMSE, and MAE to determine the best predictive model.

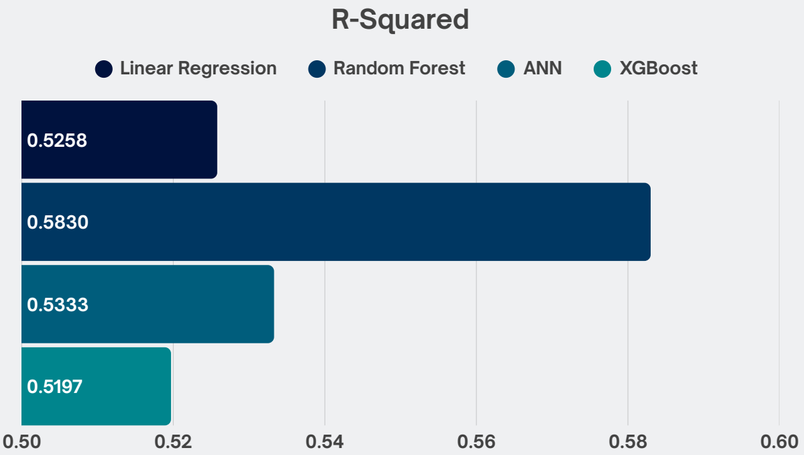
Variable Selection

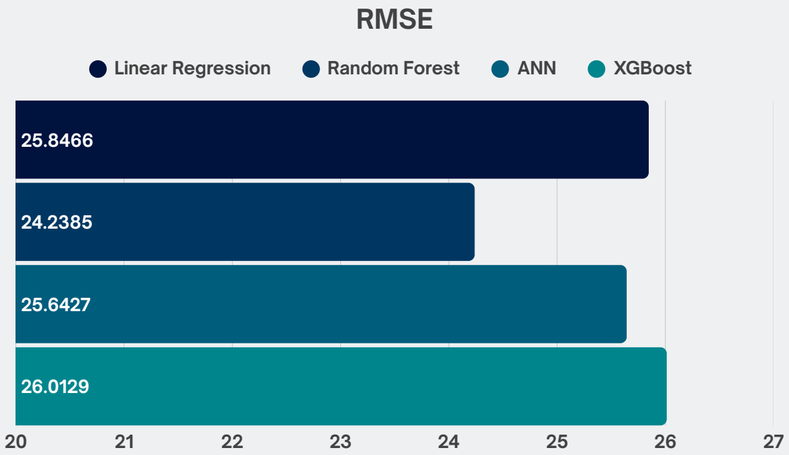
Stepwise Selection Summary						
Step	Effect Entered	Effect Removed	Number Effects In	Number Parms In	F Value	Pr > F
0	Intercept		1	1	0.00	1.0000
1	amt_bathroom		2	2	350.56	<.0001
2	size		3	3	176.58	<.0001
3	district_group		4	6	18.09	<.0001
4	type		5	7	53.77	<.0001
5	nr_school		6	8	6.54	0.0107
6	nr_skytrain		7	9	4.08	0.0435
7	nr_healthcare		8	10	5.70	0.0171

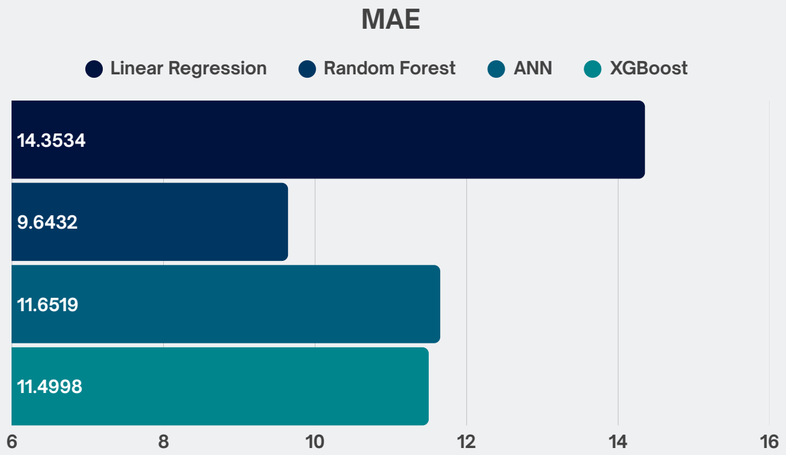
Model Performance Comparison



Model Performance Comparison

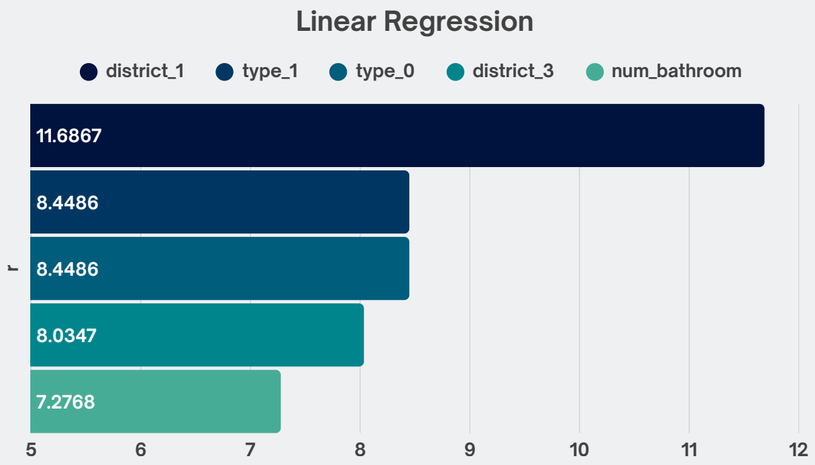


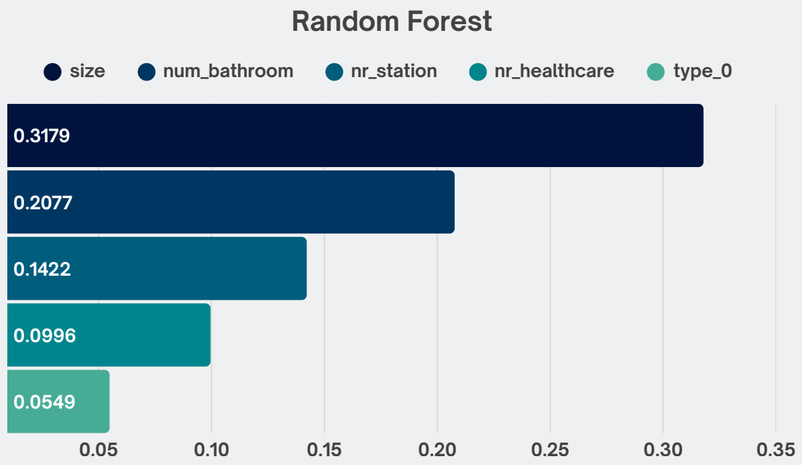


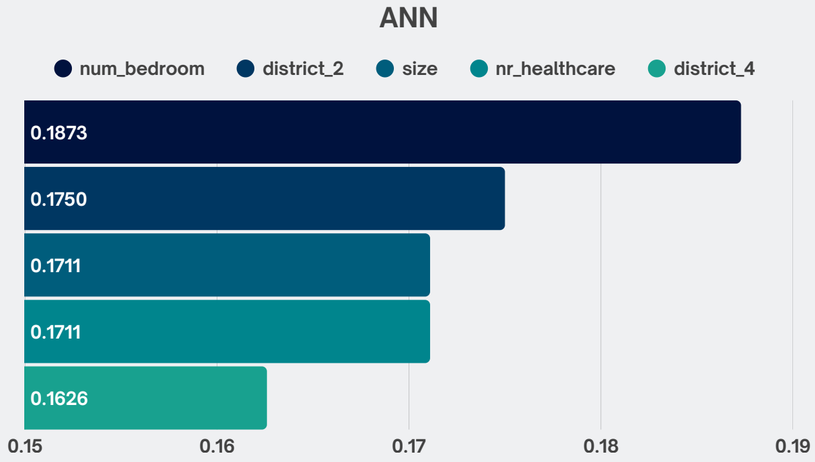


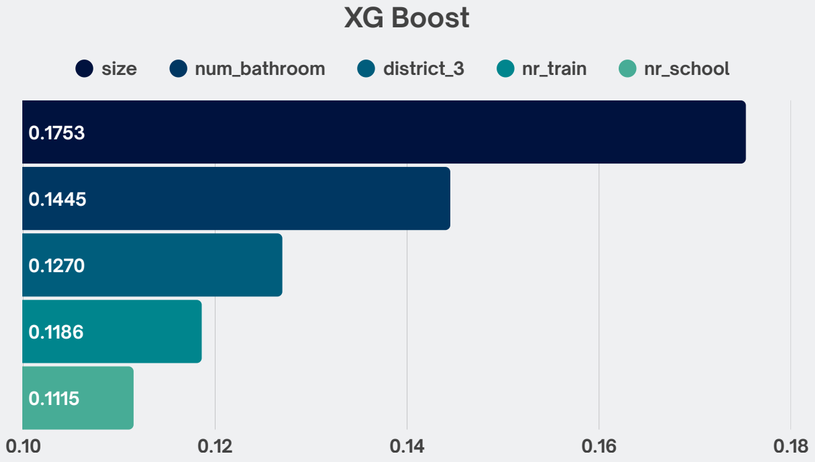
Predicting House Price in Bangkok Metropolitan

Top 5 Variables Importance









Discussions

Best Model

Random Forest shows the best performance, but R^2 is still moderate

Worst Model

XGBoost may have low performance due to untuned hyperparameters or outliers

Linear Regression

Linear Regression performs poorly due to unmet assumptions

Variable

Bedrooms, bathrooms, and house size may cause multicollinearity

Conclusion

This study evaluates predictive models for house prices in the Bangkok metropolitan area using data from 1,000 residential properties. Model performance is compared using R^2, RMSE, and MAE.

The results show that random forest performs best, with an R^2 of 0.5830, indicating moderate explanatory power and the lowest prediction errors.

The study considers factors such as house type, district, house size, number of bedrooms and bathrooms, and nearest distances to key facilities. After variable selection, the number of bedrooms, nearest distance to shopping malls, and parks are not significant.

House size is identified as the most important factor affecting house prices.

Insert your topic here

Suggestions

Data&Variables

- Distance to main roads or expressways
- Project developer
- Sales agent

Data Quality

- Multicollinearity
- Outliers

Models

- Naïve Bayes
- K-Nearest Neighbors (K-NN)
- Categorical Boosting (CatBoost)

Insert your topic here

Sources

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